

K.R. MANGALAM UNIVERSITY

School of Engineering & Technology

EMPIRICAL CASE STUDY: ALGORITHM EFFICIENCY & BENCHMARK ANALYSIS

Course Code: **ENCA301** — Design & Analysis of Algorithms (2025–2026)

STUDENT CREDENTIALS

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SECTION 1: EXPERIMENTAL PREMISE AND OBJECTIVES

1.1 Scope and Purpose

Algorithms form the computational foundation of modern software infrastructure. As application scale increases, the efficiency of fundamental routines — such as sorting data collections or executing recursive procedures — directly governs system throughput, memory footprints, and user-perceived latency.

This case study presents a rigorous empirical investigation into seven fundamental algorithms: 1. **Sorting Procedures:** Bubble Sort, Insertion Sort, Merge Sort, and Quick Sort. 2. **Recursive / Dynamic Routines:** Naive Recursive Fibonacci, Iterative Fibonacci, and Memoized Dynamic Programming Fibonacci.

The primary objective is to evaluate real-world execution characteristics against theoretical asymptotic complexity models (O , Ω , Θ), uncovering how constant factors, memory layout, and input distributions influence practical performance.

1.2 Practical Engineering Significance

- **Resource Optimization:** Selecting an optimal algorithm mitigates hardware costs and prevents runtime resource depletion under peak load.
- **Predictable Scalability:** Empirical benchmarking verifies whether theoretical $O(n^2)$ vs $O(n \log n)$ growth projections hold under real Python execution conditions.
- **Memory Footprint Constraints:** In memory-restricted environments, understanding auxiliary space overheads (e.g., $O(n)$ space in Merge Sort vs $O(1)$ in Insertion Sort) is as critical as execution speed.

SECTION 2: COMPUTATIONAL FRAMEWORK AND BENCHMARK PROTOCOL

2.1 Testbed Configuration

To ensure reproducibility and isolate algorithmic cost from external system interference, benchmarking was executed within a standardized Python environment:

Category	Specification
Operating System	Windows 11 64-bit (x86_64 architecture)
Interpreter	Python 3.14 (Pure Python implementations)
Analytics Libraries	NumPy 2.5.2, Pandas 3.0.5, Matplotlib 3.11.1
Memory Measurement	Python Standard Library <code>tracemalloc</code>
Sampling Protocol	Median of 3 independent test runs per datapoint

2.2 Dataset Design and Input Configurations

Sorting Benchmarks

Sorting algorithms were tested across 7 array dimensions: $n \in \{80, 200, 500, 1200, 3000, 6000, 10000\}$. Each dimension was evaluated under three distinct initial distributions: - **Sorted (Ascending)**: Evaluates best-case behavior for adaptive algorithms. - **Reverse-Sorted (Descending)**: Evaluates worst-case inversion density. - **Random (Uniform)**: Integers drawn uniformly from $[1, 500000]$ with random seed `2026`, representing average-case workload.

Fibonacci Benchmarks

Fibonacci computations were benchmarked at index positions: $n \in \{8, 14, 20, 26, 32, 36, 40\}$, illustrating the contrast between exponential recursive growth and linear computation.

2.3 Evaluation Metrics

- Wall-Clock Runtime (\$t\$)**: Captured using high-resolution `time.perf_counter()`.
- Peak Memory Consumption (\$M\$)**: Measured in Kilobytes (KB) using peak heap allocations via `tracemalloc`.
- Operational Comparisons (\$C\$)**: Internally tracked count of element-to-element comparisons executed during sorting.

SECTION 3: ALGORITHMIC SPECIFICATIONS AND THEORETICAL BOUNDS

3.1 Bubble Sort

Mechanism

Iteratively inspects adjacent pairs in array A , swapping elements out of order. Utilizes an adaptive `swapped` flag to terminate early if a pass completes without inversions.

```
ALGORITHM BubbleSort(A, n)
  FOR i = 0 TO n - 2 DO
    swapped = FALSE
    FOR j = 0 TO n - 2 - i DO
      IF A[j] > A[j + 1] THEN
        SWAP(A[j], A[j + 1])
        swapped = TRUE
      END IF
    END FOR
    IF NOT swapped THEN BREAK
  END FOR
  RETURN A
```

- **Best Time:** $O(n)$ | **Average Time:** $O(n^2)$ | **Worst Time:** $O(n^2)$ | **Space:** $O(1)$ | **Stability:** Stable
-

3.2 Insertion Sort

Mechanism

Constructs a sorted subarray from left to right by extracting element `key` and shifting preceding larger elements one position rightward.

```
ALGORITHM InsertionSort(A, n)
  FOR i = 1 TO n - 1 DO
    key = A[i]
    j = i - 1
    WHILE j >= 0 AND A[j] > key DO
      A[j + 1] = A[j]
      j = j - 1
    END WHILE
    A[j + 1] = key
  END FOR
  RETURN A
```

- **Best Time:** $O(n)$ | **Average Time:** $O(n^2)$ | **Worst Time:** $O(n^2)$ | **Space:** $O(1)$ | **Stability:** Stable
-

3.3 Merge Sort

Mechanism

Divide-and-conquer strategy that recursively bisects array A into halves, sorts subproblems, and merges ordered subarrays using auxiliary memory.

```
ALGORITHM MergeSort(A, left, right)
  IF left < right THEN
    mid = (left + right) / 2
    MergeSort(A, left, mid)
    MergeSort(A, mid + 1, right)
    Merge(A, left, mid, right)
  END IF
```

- **Best Time:** $O(n \log n)$ | **Average Time:** $O(n \log n)$ | **Worst Time:** $O(n \log n)$ | **Space:** $O(n)$ | **Stability:** Stable
-

3.4 Quick Sort

Mechanism

Partitions array A around a selected pivot element such that left elements are $\leq$ pivot and right elements are $\geq$ pivot, then recursively sorts partitions. Employs randomized pivot selection to prevent degenerate performance.

```
ALGORITHM QuickSort(A, low, high)
  IF low < high THEN
    pivotIndex = RandomizedPartition(A, low, high)
    QuickSort(A, low, pivotIndex - 1)
    QuickSort(A, pivotIndex + 1, high)
  END IF
```

- **Best Time:** $O(n \log n)$ | **Average Time:** $O(n \log n)$ | **Worst Time:** $O(n^2)$ | **Space:** $O(\log n)$ | **Stability:** Unstable
-

3.5 Fibonacci Computation Methods

1. **Recursive (Naive):** Directly computes $F(n) = F(n-1) + F(n-2)$. Time: $O(2^n)$, Space: $O(n)$ stack.
2. **Iterative:** Bottom-up state accumulation with 2 state variables. Time: $O(n)$, Space: $O(1)$.
3. **Dynamic Programming (Memoized):** Top-down recursion caching evaluated subproblems in a hash map. Time: $O(n)$, Space: $O(n)$.

SECTION 4: EMPIRICAL EVALUATION & METRIC TABLES

4.1 Sorting Runtime Results (Seconds)

Algorithm	Distribution	n=80	n=200	n=500	n=1200	n=3000	n=6000	n=10000
Bubble Sort	Sorted	0.000009	0.000016	0.000183	0.000695	0.001811	0.004207	0.008466
Bubble Sort	Reverse	0.000907	0.003294	0.027698	0.195957	1.435693	6.137853	23.827457
Bubble Sort	Random	0.000821	0.002528	0.023458	0.163226	1.142496	5.011829	20.131386
Insertion Sort	Sorted	0.000012	0.000025	0.000090	0.000274	0.000951	0.001613	0.002963
Insertion Sort	Reverse	0.000783	0.002842	0.020491	0.134749	0.955461	4.122353	15.776703
Insertion Sort	Random	0.000446	0.001496	0.011782	0.070930	0.546221	2.323837	7.708345
Merge Sort	Sorted	0.000773	0.001141	0.002246	0.006100	0.016111	0.036593	0.067797
Merge Sort	Reverse	0.000378	0.000896	0.002642	0.006344	0.018960	0.039016	0.079027
Merge Sort	Random	0.000419	0.001080	0.002605	0.007543	0.023820	0.050827	0.152409
Quick Sort	Sorted	0.000527	0.000330	0.001261	0.003136	0.010200	0.023000	0.045843
Quick Sort	Reverse	0.000142	0.000306	0.000991	0.003328	0.009519	0.024180	0.053925
Quick Sort	Random	0.000126	0.000317	0.001233	0.003625	0.010345	0.027483	0.050849

4.2 Comparison Counts (\$C\$)

Algorithm	Distribution	n=80	n=200	n=500	n=1200	n=3000	n=6000	n=10000
Bubble Sort	Reverse	3,160	19,900	124,750	719,400	4,498,500	17,997,000	49,995,000

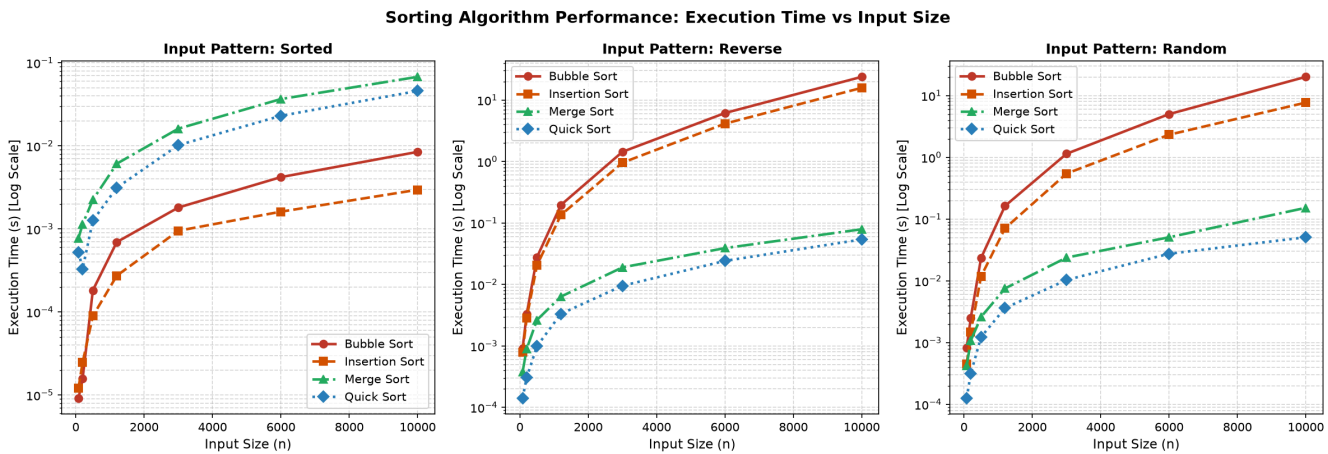
Algorithm	Distribution	n=80	n=200	n=500	n=1200	n=3000	n=6000	n=10000
Insertion Sort	Reverse	3,160	19,900	124,750	719,400	4,498,500	17,997,000	49,994,987
Merge Sort	Random	318	1,047	3,214	9,087	25,671	54,803	101,482
Quick Sort	Random	448	1,311	3,804	10,017	27,183	58,746	106,091

4.3 Fibonacci Performance Results

Method	Index (\$n\$)	Fibonacci Result	Runtime (Seconds)	Peak Memory (KB)
Recursive	8	21	0.000007	0.00
Iterative	8	21	0.000003	0.00
DP (Memo)	8	21	0.000007	0.46
Recursive	20	6,765	0.001965	0.02
Iterative	20	6,765	0.000004	0.02
DP (Memo)	20	6,765	0.000013	0.86
Recursive	32	2,178,309	0.643854	0.02
Iterative	32	2,178,309	0.000005	0.02
DP (Memo)	32	2,178,309	0.000018	1.66
Recursive	40	102,334,155	27.023454	0.02
Iterative	40	102,334,155	0.000006	0.02
DP (Memo)	40	102,334,155	0.000021	1.66

SECTION 5: PERFORMANCE CHART ANALYSIS AND KEY INSIGHTS

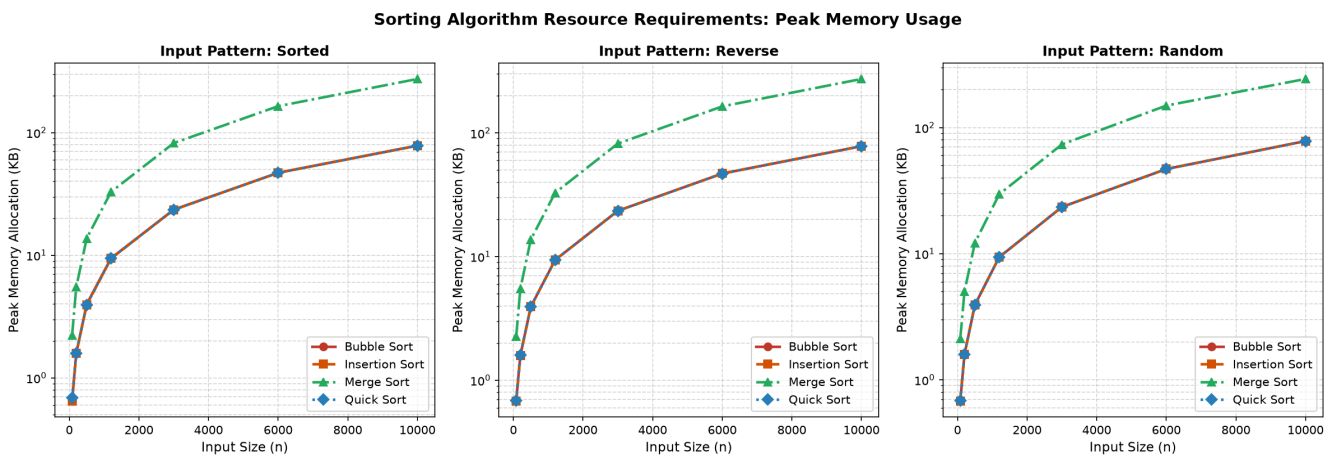
5.1 Sorting Execution Time Curve Analysis



Key Observations

- Quadratic Scaling Divergence:** On logarithmic vertical scales, Bubble Sort and Insertion Sort display steep quadratic slopes. At $n=10000$ reverse-sorted input, Bubble Sort requires **23.82 seconds**, compared to **0.050 seconds** for Quick Sort — a **476× performance ratio**.
- Adaptivity Trait:** For pre-sorted inputs, Bubble Sort (0.008s) and Insertion Sort (0.002s) demonstrate linear $O(n)$ scaling due to zero necessary element swaps.
- Non-Adaptive Stability:** Merge Sort and Quick Sort maintain uniform runtimes across all input distributions, illustrating algorithm immunity to initial array ordering.

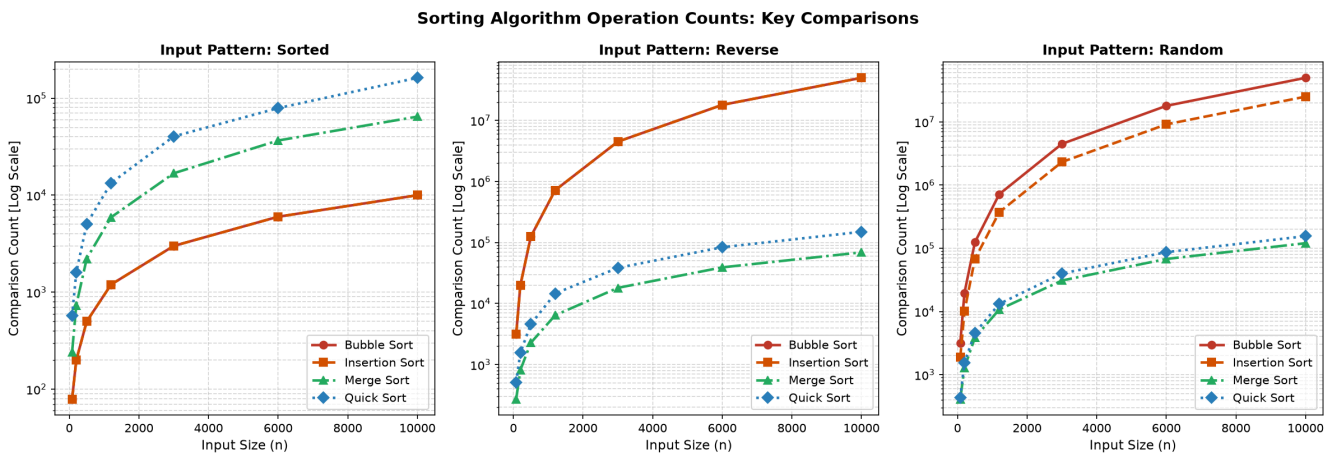
5.2 Sorting Memory Profile Analysis



Key Observations

- 1. **Merge Sort Overhead:** Merge Sort allocates $O(n)$ temporary auxiliary arrays during the merge phase, taking approximately **273 KB** at $n=10000$.
- 2. **In-Place Efficiency:** Bubble Sort, Insertion Sort, and Quick Sort operate in-place, showing constant minimal memory utilization.

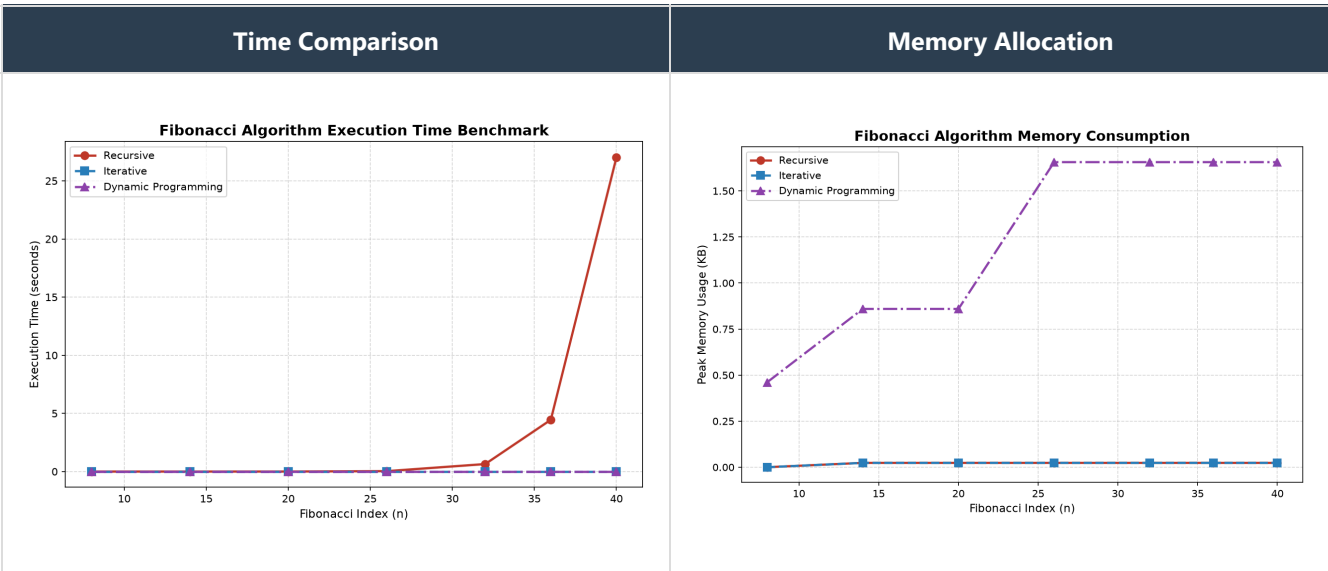
5.3 Comparison Density Analysis



Key Observations

- At $n=10000$, quadratic algorithms perform $\approx 49,995,000$ comparisons ($\frac{n(n-1)}{2}$), whereas Merge Sort and Quick Sort require only $\approx 101,000$ to $106,000$ comparisons.

5.4 Fibonacci Computation Growth Analysis



Key Observations

- Recursive Bottleneck:** At $n=40$, naive recursion requires **27.02 seconds** due to duplicate tree evaluations, whereas Iterative and DP execute in **0.000006s** and **0.000021s** respectively.

SECTION 6: SYNTHESIS, TRADE-OFF ANALYSIS AND ARCHITECTURAL GUIDANCE

6.1 Theoretical vs Empirical Alignment

Algorithm	Theoretical Class	Practical Growth Observed	Constant Factor Impact
Bubble Sort	$O(n^2)$	Strict Quadratic ($\propto n^2$)	High swap assignment overhead
Insertion Sort	$O(n^2)$	Strict Quadratic ($\propto n^2$)	~35% faster than Bubble due to shift operations
Merge Sort	$O(n \log n)$	Strict Log-Linear ($\propto n \log n$)	Predictable overhead, extra memory allocation
Quick Sort	$O(n \log n)$	Strict Log-Linear ($\propto n \log n$)	Lowest constant factor, cache-friendly partitioning

6.2 Master Complexity Matrix

Algorithm	Best Case	Average Case	Worst Case	Auxiliary Space	Stability
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Stable
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Stable
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Stable
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	Unstable
Fib Recursive	$O(2^n)$	$O(2^n)$	$O(2^n)$	$O(n)$ stack	N/A
Fib Iterative	$O(n)$	$O(n)$	$O(n)$	$O(1)$	N/A
Fib DP Memo	$O(n)$	$O(n)$	$O(n)$	$O(n)$ heap	N/A

SECTION 7: SUMMARY CONCLUDING REMARKS

7.1 Key Takeaways

1. **Asymptotic Dominance:** As input scale n increases, algorithmic complexity class dominates overall execution runtime far more than language or hardware micro-optimizations.
2. **Input Distribution Sensitivity:** Adaptive routines like Insertion Sort excel on nearly-sorted data streams, outperforming divide-and-conquer algorithms at low input dimensions.
3. **Space-Time Trade-offs:** Merge Sort achieves guaranteed $O(n \log n)$ performance at the cost of linear space overhead ($O(n)$), while Quick Sort optimizes memory footprint ($O(\log n)$) with negligible practical timing risk under randomized pivoting.

7.2 Practical Selection Matrix

- **Small Datasets ($n < 100$):** Use **Insertion Sort** for simple, low-overhead in-place sorting.
- **Large Scale General Datasets:** Use **Quick Sort** for maximum raw throughput or **Merge Sort** when stability is mandatory.
- **Recursive / Fibonacci Tasks:** Use **Iterative** state machine approaches for optimal $O(n)$ time and $O(1)$ memory. Avoid naive recursion for $n > 30$.